LESSON PLAN

AUXILIARY EQUIPMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Classification of bearing
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Classification of bearing
5. Specific Objectives	18.1.3.1 Trainee will be able to explain operation & functions of bearing

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of bearing types from IB/PDF; Instructor supervision required for diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a bearing?
(b) Why are bearings essential in aircraft auxiliary equipment?
(c) Can you name one type of bearing used in engines?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and basic operation of bearings • Classification by friction type • Classification by rolling element shape • Classification by direction of load • Classification by housing design	L & Ds	PPT, IB	15 min
(b)	• Deep groove ball bearings characteristics • Tapered roller bearings load capacity • Needle roller bearings space efficiency • Thrust bearings axial load handling • Sleeve bearings lubrication requirements	L & Ds	PPT, PDF	20 min
(c)	• Selection criteria for auxiliary equipment • Maintenance implications of bearing types • Failure modes and identification • Replacement procedures overview • Safety precautions during inspection	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between radial and thrust bearings.
(b) How does the rolling element shape affect bearing selection?
(c) What are the key maintenance considerations for sleeve bearings?

12. Home Assignment and Follow Up	Prepare a brief note on the specific bearing types used in the aircraft auxiliary equipment covered in LP 02, highlighting their operational functions.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Checks on bearing	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min